



On Cu^{2+} , Zn^{2+} and B^{3+} cations substitution in polymer sol-gel derived YAG garnets: Catalytic behavior in ethylbenzene oxidation reaction

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ABSTRACT

The incorporation of host cations B^{3+} , Cu^{2+} or Zn^{2+} into the yttrium aluminum garnet (YAG) lattice was performed by polymer-assisted sol-gel method in an acid medium or in its absence. XRD, FTIR and Raman spectroscopy revealed $\text{Y}_3\text{Al}_5\text{O}_{12}$, YAlO_3 , $\text{Y}_4\text{Al}_2\text{O}_9$ phases in all solids, independently of the synthesis route, besides the $\text{Y}_2\text{Cu}_2\text{O}_5$ phase for Cu-based solids. The micro-mesoporous structure of the solids, possessing small sized particles was modulated by preparing the samples in acid media, with most of host cation Zn^{2+} and Cu^{2+} entering the YAG lattice. XPS and EPR results illustrated a large number of oxygen vacancy defects along with the presence of Cu^{2+} and Cu^+ surface species in Cu-based samples, which contributed to good catalytic behavior in the ethylbenzene oxidation reaction to produce acetophenone and benzaldehyde. The structural stability of a particular solid containing surface $\text{Y}_2\text{Cu}_2\text{O}_5$ phase was demonstrated within four times of reuse with 40 % ethylbenzene conversion and selectivity for acetophenone of 34 %.

1. Introduction

Yttrium aluminum garnet (YAG) is an important class of transparent polycrystalline ceramic material that possesses good thermochemical stability and remarkable physicochemical properties [1,2].

Specifically, the perfect YAG garnet-type structure is known to exhibit a cubic close-packed structure with a unit cell of 160 atoms, e.g., a primitive cell of 80 atoms, holding a cell and lattice constant of 12.000 Å ($Ia\bar{3}d$ (O_h^{10}) space group) [1–4]. Such a structure possesses eight formula units consisting of $\text{A}_3\text{B}_2\text{B}''_3\text{O}_{12}$, where A, B' and B'' are eight-fold coordinated in a distorted cubic D_2 local site, with cation sites of octahedral and tetrahedral symmetry [4]. The crystal structure can be built from position A with Y^{3+} ions in 24(c) sites (0.1109 nm) occupying dodecahedral coordination, whereas B' and B'' positions can be Al^{3+} ions existing at 16(a) sites (0.0535 nm) and 24(d) sites (0.039 nm) crystallographically coordinated [5,6]. Besides, this highly stable polyhedral framework is corner-shared with oxygen atoms existing at 96 (h) sites, in

which each is assembled with two Y^{3+} corner-sharing dodecahedra, one Al^{3+} tetrahedron and one Al^{3+} octahedron [4–6], as shown in Fig. 1. All the tetrahedra and octahedra are isolated from each other, but form a stable structure. The YAG garnet-type structure has the formula $\text{Y}_3\text{Al}_5\text{O}_{12}$, which remains unaffected after the cations substitution in the lattice or the doping different atoms at several sites in the garnet [5,6].

Currently, several methods have been used for the synthesis of YAG garnets, including solid-state reaction between alumina and yttrium oxides, wet chemical methods such as sol-gel process, polymerized complex route, microwave irradiation, solvothermal, microemulsion synthesis of yttrium carboxylate and co-precipitation from solutions to obtain YAG phases [7–9]. Compared with wet chemical methods, sol-gel synthesis and co-precipitation often result in controllable morphology and porosity, generate defects, and adjust particle size and regulate the phase purity composition, depending on the annealing temperature [9–11].

When materials are fabricated in a nanostructured dimension, the

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